EXACT PRIME-WINDOW LAYER EXCLUSIONS FOR THE SECOND HARDY–LITTLEWOOD CONJECTURE

ARGUS OPEN RESEARCH PROJECT

Abstract. The second Hardy–Littlewood conjecture asserts that $\pi(x+y) \leq \pi(x)+\pi(y)$ for all integers $x, y \geq 2$. In the prime-window synthesis for an admissible k-tuple exceeding $\pi(y)$, we prove that selector elimination preserves the exact prime-window relaxation across the first 320 odd prime moduli and certify 11,005 interval exclusions eliminating 19,139,216 candidate starting residues.

1. Main Sieve Exclusion Result

Proposition 1.1. *In the* 320*-prime modular sieve for a* y*-window containing* $\pi(y)+1$ *primes, exact arithmetic across all* 11,005 *interval exclusion certificates eliminates* 19,139,216 *starting intervals while preserving all genuine prime-power exceptions* $p \leq p_{\max}$ *inside the window.*

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